



Fig. 7(A): The output waveform of 'forward ramp wave'

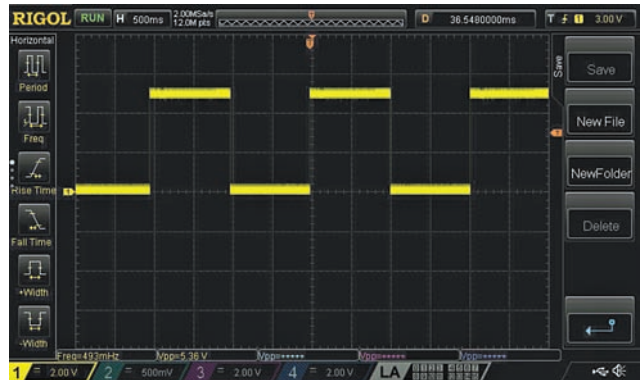


Fig. 7(D): The output waveform of 'square wave'



Fig. 7(B): The output waveform of 'reverse ramp wave'



Fig. 7(E): The output waveform of 'triangular wave'

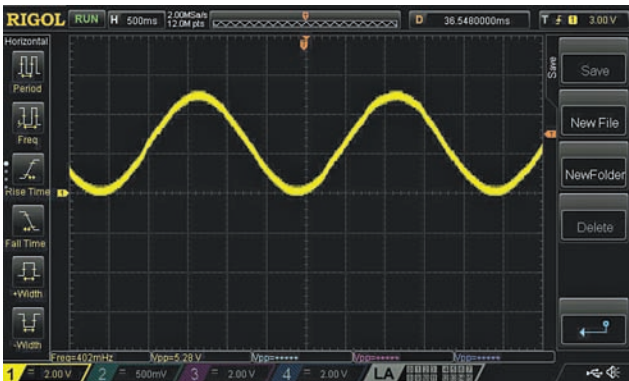


Fig. 7 (C): The output waveform of 'sine wave'

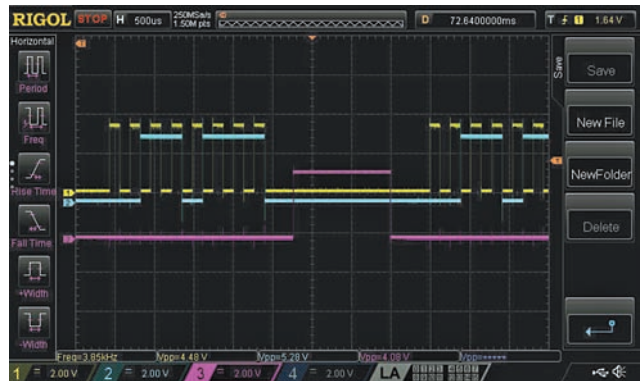


Fig. 7 (F): SPI serial communication from Pi to the waveform generator: yellow trace – SCLK, blue trace – SDATA, purple trace – strobe/latch

it. Alternatively, open the folder in the terminal and use the following command:

```
sudo python3 "your pythonfilename".py
```

For example, if my sine wave file name is sine.py then I run

```
sudo python3 sine.py
```

Construction and testing

Assemble the circuit on the PCB.

Fig. 4 shows the completely

assembled board for the waveform generator device. The circuit was assembled on a professional-quality PCB, as shown in Fig. 4. Initially, only the R-2R DAC section can be assembled, and the output behaviour of the DAC should be tested for various logic data inputs ranging from 0 to 255 in binary format, such as 00000000 to 11111111. This is a

time-consuming process, but testing the DAC for its linearity also ensures proper waveform generation at a later stage. Additionally, this testing saves time in the event of any issues with the waveform generator.

Fig. 5 shows the assembled board used during testing, while Fig. 6 illustrates the linearity of the DAC for data inputs from 0 to 255.